CONCEPT 43.1

In innate immunity, recognition and response rely on traits common to groups of pathogens

For a **pathogen**—a bacterium, fungus, virus, or other disease-causing agent—the internal environment of an animal offers a source of nutrients, a protected setting, and a means of transport to new environments. From the animal's vantage point, the situation is not so ideal. Fortunately, adaptations have arisen over the course of evolution that protect animals against many pathogens. The body's defenses make up the **immune system**, which enables an animal to avoid or limit many infections. (Note that a foreign molecule or cell doesn't have to be pathogenic to elicit an immune response, but we'll focus on the immune system's role in defending against pathogens.)

The first lines of defense offered by immune systems help prevent pathogens from gaining entrance to the body. For example, an outer covering, such as skin or a shell, blocks entry by many pathogens. Sealing off the entire body surface is impossible, however, because gas exchange, nutrition, and reproduction require openings to the environment. Secretions that trap or kill pathogens guard the body's entrances and exits, while the linings of the digestive tract, airway, and other exchange surfaces provide additional barriers to infection.

If a pathogen breaches barrier defenses and enters the body, the problem of how to fend off attack changes substantially. Housed within body fluids and tissues, the invader is no longer an outsider. To fight infections, an animal's immune system must detect foreign particles and cells within the body. In other words, a properly functioning immune system distinguishes nonself from self. How is this accomplished? Immune cells produce receptor molecules that bind specifically to molecules from foreign cells or viruses and activate defense responses. The specific binding of immune receptors to foreign molecules is a type of *molecular recognition* and is the central event in identifying nonself molecules, particles, and cells.

Two types of immune defenses are found among animals. This concept focuses on **innate immunity**, the set of immune defenses common to all animals. The remainder of the chapter explores **adaptive immunity**, a set of molecular and cellular defense found only among vertebrates.

Innate Immunity of Invertebrates

The great success of insects in terrestrial and freshwater habitats teeming with diverse pathogens highlights the effectiveness of invertebrate innate immunity. One part of this defense system is a set of barrier defenses, including the insect exoskeleton. Composed largely of the polysaccharide chitin, the exoskeleton provides a physical barrier against most pathogens. Chitin also lines the insect intestine, where it blocks infection by many pathogens. In the digestive system, **lysozyme**, an enzyme that breaks down bacterial cell walls, acts as a chemical barrier against any pathogens ingested with food.

Any pathogen that breaches an insect's barrier defenses encounters internal immune defenses. Insect immune cells produce a set of recognition proteins, each of which binds to a molecule common to a broad class of pathogens. Many of these molecules are components of fungal or bacterial cell walls. Because such molecules are not normally found in animal cells, they function as "identity tags" for pathogen recognition. Once bound to a pathogen molecule, a recognition protein triggers an innate immune response.

In insects, the major immune cells are called *hemocytes*. Like amoebas, some hemocytes are phagocytic cells: They ingest and break down microorganisms by a process known as **phagocytosis** (**Figure 43.2**). One class of hemocytes produces a type of defense molecule that helps entrap larger pathogens, such as *Plasmodium*, the single-celled parasite of mosquitoes that causes malaria in humans. Many other hemocytes release *antimicrobial peptides*, which circulate throughout the body of the insect and inactivate or kill bacteria or fungi by disrupting their plasma membranes.

The innate immune response of insects is specific for particular classes of pathogens. For example, if a fungus infects an insect, binding of recognition proteins to fungal cell wall molecules activates a transmembrane receptor called Toll. Toll in turn activates production and secretion of antimicrobial peptides that specifically kill fungal cells. Remarkably,

▼ **Figure 43.2 Phagocytosis.** This diagram depicts events in the ingestion and destruction of pathogens by a typical phagocytic cell.